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PROFESSIONAL PROFILE

Accomplished Software Engineer and Data Analyst with three years of academic project experience in multiple data management and software development projects. Experience includes:

- Collected and analyzed user requirements to write and test code.
- Developed existing programs by refactoring and improving the pre-written code.
- Developed and implemented data analyses, data collection systems, and other strategies to optimize statistical efficiency and quality; participated in scrum teams and agile delivery to meet sprint goals.
- Utilized open-source technologies to mine and collect various data sources and platforms.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, R, Kotlin
Markup: Markdown, LaTeX
Tools: MATLAB, SPSS, Jupyter Notebook, Unity, M.S. Visual Studio, Git, Valgrind, GNU Make
Libraries: NumPy, Matplotlib, pandas
Operating Systems: Windows, Linux, MacOS
Design: Adobe Photoshop, Adobe Lightroom

EDUCATION

Virginia Polytechnic Institute and State University, Blacksburg, VA

Master of Engineering: Computer Science *– Anticipated May 2022*
Bachelor of Science: Computational Modeling and Data Analytics *– Dec 2020*
Minor in Computer Science, Mathematics

RELEVANT ACADEMIC PROJECTS

Virginia Tech **Jan 2020 – Present**

Research Collaborator - LED Block Cipher in Securing Healthcare Data*, Falls Church, VA (Oct 2021 – Dec 2021)*

- *Information Security Class*: Working with three classmates, researched the background of lightweight cryptography in the healthcare industry and the Light Encryption Device (or LED) algorithm being incorporated with IoT devices to secure healthcare data.
- Wrote the code for the LED algorithm using Python and experimented with different choices of healthcare data using PyCham and Visual Code software to record its running time and space usage in RAM. Executed the algorithm along with SIMON, SPECK, and AES algorithms to compare efficiencies.
- *Projected Results*: LED was a good algorithm to use for devices with limited capacity due to constant space usage, but was not as good as the other algorithms in terms of time efficiency.

Program Devloper - COVID-19 Social Project*, Blacksburg, VA (Apr 2020 – May 2020)*

- *Academy of Integrated Science*: Explored the outcomes/influences of the pandemic on local neighborhoods with varying income levels to discover healthcare and healthy food access. Cleaned the data using Python
- Utilized quantitative and qualitative data analysis, exploratory analysis, and cluster analysis methodologies, MATLAB, ggplot2, and Matplotlib tools, and Jupyter Notebook, PyCharm, and RStudio software.
- Metrics: accuracy and precision of models, F1-score, entropy
- *Results*: Discovered that K-Nearest Neighbors and Logistic Regression were the best candidates to classify the dataset in comparison to Decision Tree and Naive Bayes algorithms. Found that families living in lower-income and higher-income areas had adequate access to healthy food and pharmacies but still susceptible to health risks.

Data Analyst, DataScope's Six Sigma*, Blacksburg, VA (Jan 2020 – May 2020)*

- *Calhoun Discovery Program*: Obtained a 25MB file of data from a factory using Quadbots and dissected the given dataset into four smaller chunks. Utilized exploratory data analysis in R to statistically analyze the number of defective parts and manufacturing time.
- *Results*: Detected four Quadbots that manufactured the highest number of defective components in minimal time by implementing an algorithm that utilized Six Sigma in Python.

PROFESSIONAL EXPERIENCE

Program Developer, *Sign Language Reader*, Glen Allen, VA Jan 2022 – Present

- Studied and researched different gestures in sign language.
- Brainstormed and created diagrams of a system that uses machine learning algorithms to convert sign language to written language.